

QuoteZen Engineering Investigation

Quote #548: Services Calculation Root Cause, Fix Implementation & Price Alignment

QUOTE: #548 (Storefront) PRODUCT: LEDFul ISD320 / IF1.5-160 (1120x1920mm) STATUS: **FIXED & RECOMPUTED (14 hrs / \$2,978 Services / \$11,907 Total)**

1. Executive Summary: Problem, Solution & Verification

Why Services calculated at \$14,891, how it was corrected to \$2,978, and origin of the \$475 figure

Key Root Cause & Permanent Fix

Root Cause (Module vs Cabinet Ingestion): In the Excel reference sheet, LED products define standard physical cabinets in Cols E & F (960x960mm) and module sizes in Cols L & M (160x160mm). The original catalog ingestion script only captured Cols L & M as minCabinet dimensions. For an 1120x1920mm screen, QuoteZen treated every 160x160mm module as an individual cabinet, computing 84 cabinets (7x12) and 90 installer hours (\$8,550 cost) instead of 4 standard cabinets (2x2) and 14 installer hours (\$1,330 cost) — inflating labour by 76 hours (\$7,220 cost / \$11,913 sell).

Permanent Fix Implemented: Added cabinet_w_mm and cabinet_h_mm to the LedProduct PostgreSQL model. Updated extract_catalog.py to ingest Cols E & F and synchronized all 176 LED products. Updated computeLedScreenPricing() to calculate installer labour using standard cabinet dimensions while retaining module dimensions for precision screen snapping.

Origin of the \$475 Figure (Standard Overhead, Not Sea Freight): The \$475 allowance is NOT sea freight. It is the Standard Overhead Allowance defined in Reference Data / settings table (composed of \$120 PM, \$80 design/site prep, \$80 screen configuration, \$75 warehouse, \$60 rubbish disposal, \$30 consumables, and \$30 controller courier). Screen delivery freight was \$0 because freightOptionId was not selected.

Explanation of the Remaining ~\$370 Difference (\$2,978.25 vs ~\$2,607 in Excel): 1) Commercial Multiplier: QuoteZen applies the standard commercial Service Markup of 1.65x (\$1,805 cost x 1.65 = \$2,978.25 sell). In contrast, this specific client workbook used the Client Margin Divisor formula: Sell = Cost / (1 - 0.29) = \$1,805 / 0.71 = \$2,542 sell (1.408x effective multiplier). 2) Overhead Allocation: In multi-screen projects (like Baby Bunting), the estimator halved overhead quantities to 0.5 (\$337.50 cost vs \$475 standard) because PM and warehouse costs are shared across all screens.

2. Quote #548 Services Breakdown: Before vs After Resolution

Item-by-item breakdown of installer hours, overhead components, costs, and final sell prices

Component / Service Item	Before Fix (84 Cabs)	After Fix (4 Cabs)	Cost (AUD)	Sell (1.65x)
Installer Labour (Total)	90.0 hrs (\$8,550 cost)	14.0 hrs (\$1,330 cost)	\$1,330.00	\$2,194.50
• Size-driven cabinet hours	84.0 hrs (84 modules x 1.0)	8.0 hrs (4 cabs x 2.0)	\$760.00	\$1,254.00
• Base install allowance	4.0 hrs base	4.0 hrs base	\$380.00	\$627.00
• Frame install allowance	2.0 hrs (ivisual Low Frame)	2.0 hrs (ivisual Low Frame)	\$190.00	\$313.50
• Location hourly uplift	\$0 (VIC Metro)	\$0 (VIC Metro)	\$0.00	\$0.00
Standard Overhead Allowance	\$475.00 cost	\$475.00 cost	\$475.00	\$783.75
• Project Management (PM)	1.0 allowance (\$120)	1.0 allowance (\$120)	\$120.00	\$198.00
• Installation Design & Prep	1.0 allowance (\$80)	1.0 allowance (\$80)	\$80.00	\$132.00
• Screen Configuration	1.0 allowance (\$80)	1.0 allowance (\$80)	\$80.00	\$132.00
• Warehouse Handling	1.0 allowance (\$75)	1.0 allowance (\$75)	\$75.00	\$123.75
• Rubbish Disposal	1.0 allowance (\$60)	1.0 allowance (\$60)	\$60.00	\$99.00
• Consumables & Fasteners	1.0 allowance (\$30)	1.0 allowance (\$30)	\$30.00	\$49.50
• Controller Courier / Delivery	1.0 courier (\$30)	1.0 courier (\$30)	\$30.00	\$49.50
Freight / Access / Engineering	None selected (\$0)	None selected (\$0)	\$0.00	\$0.00
TOTAL SERVICES (Quote #548)	90.0 hrs / \$14,891.25	14.0 hrs / \$2,978.25	\$1,805.00	\$2,978.25

3. Side-by-Side Comparison: Excel Formula Trace vs QuoteZen

Technical comparison of geometry, labour estimation, pricing math, and final figures

Dimension / Parameter	Excel Workbook (2026-XXX Quote Base)	QuoteZen (After Implemented Fix)
Cabinet Source Data	Cols E & F: standard 960 x 960 mm Calc L & M: module 160 x 160 mm	LedProduct.cabinetWmm = 960 mm LedProduct.minCabinetWmm = 160 mm
Screen Snapping vs Labour	Snapping: 7x12 modules = 1120x1920mm Labour: 2x2 standard cabinets = 4 cabs	Snapping: 7x12 modules = 1120x1920mm Labour: 2x2 standard cabinets = 4 cabs
Cabinet Size Factor	960 x 960 / 1000 = 921.6 > 600 -> factor = 2.0	960 x 960 / 1000 = 921.6 > 600 -> factor = 2.0
Size-Driven Labour Hours	MAX(4 cabinets x 2.0, 4) = 8.0 hours	MAX(4 cabinets x 2.0, 4) = 8.0 hours
Base & Frame Allowance	4 hrs base + 2 hrs frame = 6.0 hours	4 hrs base + 2 hrs frame = 6.0 hours
Total Labour Hours	14.0 hours	14.0 hours (exact match)
Labour Cost (@ \$95/hr)	14 hrs x \$95 = \$1,330.00	14 hrs x \$95 = \$1,330.00 (exact match)
Overhead Allowance Cost	\$337.50 (halved for multi-screen) or \$475 std	\$475.00 (standard single-screen package)
Total Services Cost	\$1,667.50 (halved) / \$1,805.00 (standard)	\$1,805.00 (exact standard cost match)
Pricing Sell Method	Client Margin Divisor: Cost / (1 - 0.29) or (1 - 0.30)	Service Commercial Markup: Cost x 1.65
Final Services Sell Result	~\$2,607.00 (Sheet L2 proposal)	\$2,978.25 (stored quoteServices)
Quote Grand Total	~\$11,752.00	\$11,907.86 (within 1.3% variance)

4. Engineering Implementation & Reference Data Origins

Summary of codebase changes, database migrations, and reference data sources

Code & Database Changes Implemented

- Database Schema & Migration:** Added cabinet_w_mm and cabinet_h_mm (Decimal(10,2)) to the LedProduct model in Prisma. Applied migration 20261003130000_led_product_cabinet_dimensions to RDS PostgreSQL via prisma migrate deploy.
- Catalog Extraction & Seeder Sync:** Updated extract_catalog.py to extract Cols E & F into catalog.json. Updated import-catalogs.ts to sync standard dimensions on all 176 LED products. Product 10 (ISD320 / IF1.5-160) now stores minCabinet: 160x160mm, cabinet: 960x960mm.
- Labour Calculation Alignment:** In computeLedScreenPricing() (apps/api/src/modules/quotes/screens.ts), derived cabinetsW and cabinetsH from standard cabinet dimensions (with rotateCabinets support), eliminating module-level over-counting.

Where Does the \$475 Standard Overhead Come From?

- Workbook Reference Data Source:** In Reference Data and sheet (LED 1) rows 294-309, Seen Technology defines standard overhead allowances: Consumables (\$30), Rubbish (\$60), Warehouse (\$75), Project Management (\$120), Installation Design (\$80), Screen Config (\$80), Courier (\$30). Sum = \$475.00.
- QuoteZen Configuration & Database Settings:** In QuoteZen, WORKBOOK_DEFAULTS.freight.standardOverheadAud defaults to 475. It is dynamically loaded from the settings table in PostgreSQL under key "standard_overhead", allowing admins to adjust it globally without code changes.